

Applied Bioinformatics Laboratory

Experiment – 6: Pharmacogenomics Data Analysis using PharmGKB

Name: Jayasuriya J

AIM:

To investigate how genetic differences influence drug response using the PharmGKB database, taking omeprazole as an example drug.

INTRODUCTION:

Pharmacogenomics examines the role of genetic variability in determining how individuals respond to medications. Differences in genes that encode drug-metabolizing enzymes, transport proteins, or drug targets can result in variations in drug effectiveness, toxicity, and optimal dosage. PharmGKB (Pharmacogenomics Knowledgebase) is a comprehensive, curated database that compiles information on gene–drug relationships, genetic variant annotations, drug labeling details, and clinical practice guidelines. It serves as a valuable resource for understanding the genetic basis of drug response. Omeprazole, a commonly prescribed proton pump inhibitor for acid-related gastrointestinal conditions, is mainly metabolized by the CYP2C19 enzyme. Polymorphisms in the CYP2C19 gene play a crucial role in altering the metabolism of omeprazole, leading to differences in therapeutic response among individuals.

METHODOLOGY:

1. The PharmGKB website was opened and accessed.
2. The term “omeprazole” was entered into the search function.
3. The Drug Label Annotations section was examined to identify pharmacogenomic information provided by regulatory authorities.
4. The Variant Annotations section was explored to assess the relationship between specific genetic variants and drug response.
5. Genes and genetic variants associated with omeprazole were recorded, focusing primarily on CYP2C19.
6. Relevant scientific publications (PMIDs) linked to the annotations were reviewed to evaluate supporting evidence.

QUESTION:

To study the effect of genetic variation on drug response using the PharmGKB database.

RESULTS:



Fig 1: ClinPGx – Evolution of PharmGKB Homepage



Fig 1.2: Omeprazole Overview Page in PharmGKB

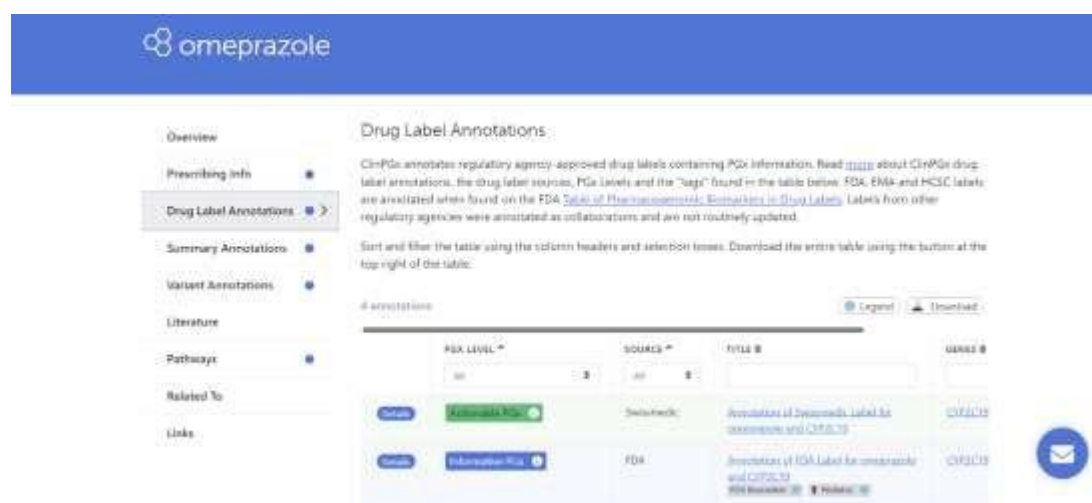


Fig 1.3: Drug Label Annotations for Omeprazole

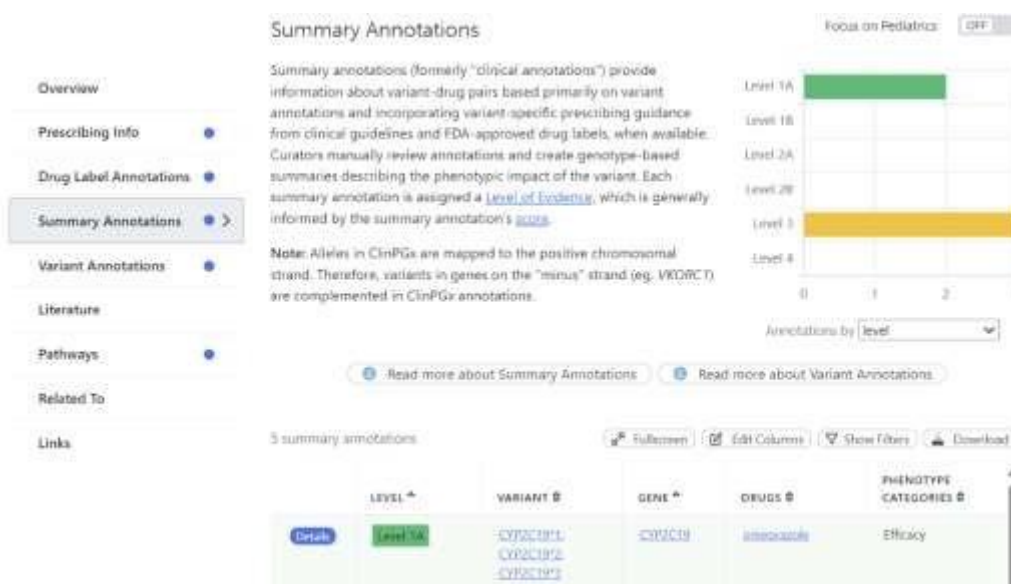


Fig 1. 4: Summary Annotations of Omeprazole

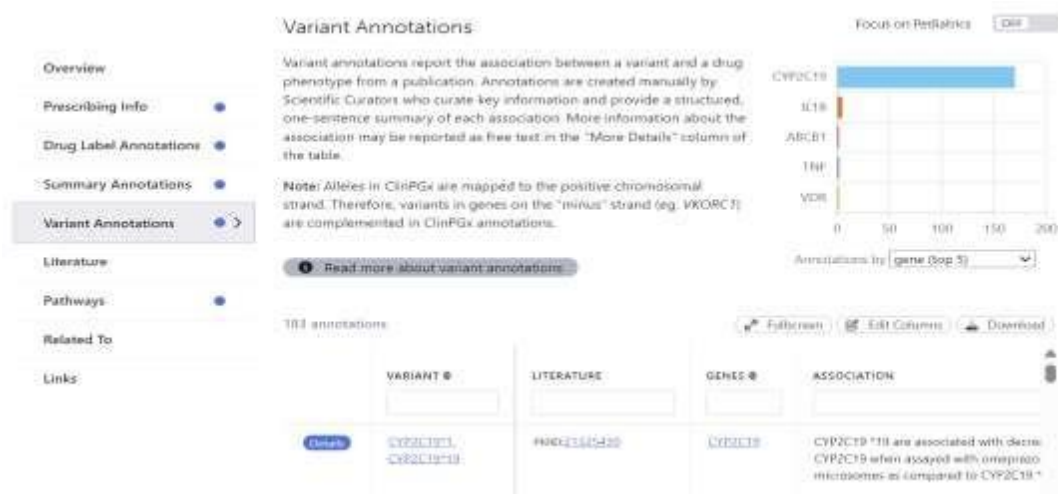


Fig 1.5: Variant Annotations of Omeprazole

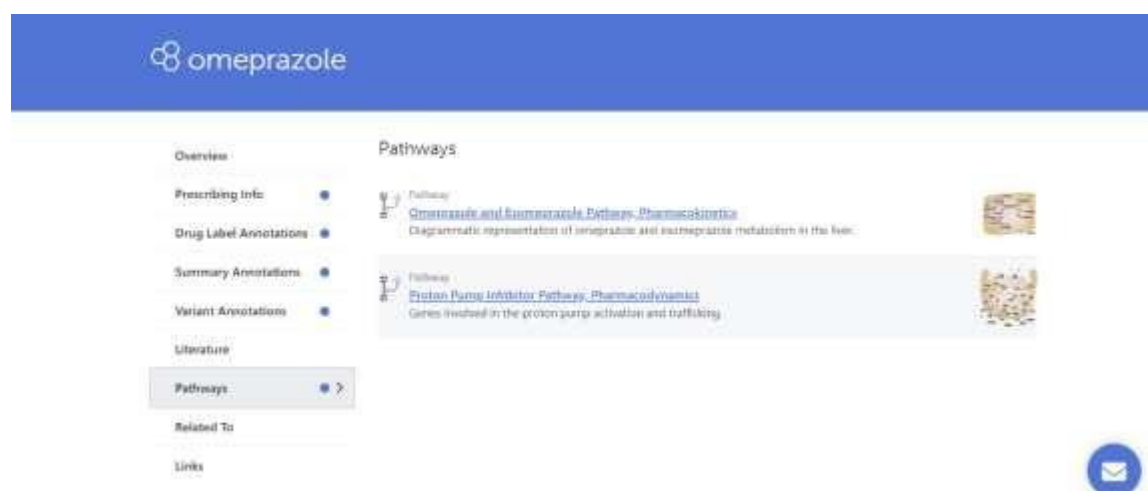


Fig 1.6: Omeprazole-Associated Pathways in PharmGKB

INTERPRETATION:

The Drug Label Annotations for omeprazole indicate clinically relevant pharmacogenomic information, particularly involving the CYP2C19 gene. Regulatory agencies such as the FDA and Swissmedic provide pharmacogenomic labeling for omeprazole, highlighting the importance of CYP2C19 variants. The Variant Annotations section shows a high number of annotations linked to CYP2C19, indicating it as the major gene influencing omeprazole metabolism. Variants such as CYP2C19*1 and CYP2C19*19 are associated with altered enzyme activity. Individuals carrying reduced-function or loss-of-function CYP2C19 alleles show decreased metabolism of omeprazole, which can result in higher plasma drug levels. Other genes such as IL1B, ABCB1, TNF, and VDR show minor associations, but their impact is less significant compared to CYP2C19. These findings demonstrate that genetic variation directly affects drug response, supporting the need for genotype-guided therapy.